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CENG 407 - Implementation Details

VR Anatomy: VR Based Educational Interactive Human Anatomy Training Platform

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1. Introduction

The rapid advancement of virtual reality (VR) technologies has opened new opportunities in educational environments, particularly in fields that require strong visual and spatial understanding such as human anatomy. Traditional anatomy education methods are often limited in terms of interactivity and student engagement. Therefore, integrating immersive technologies like VR can significantly enhance the learning experience.

This project, titled “VR Anatomy: VR Based Educational Interactive Human Anatomy Training Platform”, aims to develop an interactive virtual reality application that enables users to explore human anatomy in a more engaging and intuitive way. The system allows users to interact with 3D anatomical models, examine structures closely, and access relevant information in real time.

The development process of the project is divided into two main phases: CENG 407 and CENG 408. In CENG 407, a basic working prototype was developed, focusing on core VR interactions and simple model visualization. In CENG 408, the project was expanded into a full-featured application by integrating multiple modules such as interactive learning modes, free exploration, quiz systems, and AI-assisted interaction.

This report focuses on the implementation details of the project in CENG 408, including the developed features, iteration process, system improvements, and technical components used throughout the development lifecycle.

2. System Overview

The VR Anatomy system is developed using Unity 6 as the main development platform and is designed to run on Meta Quest 2 virtual reality devices. The system provides an immersive and interactive learning environment where users can explore human anatomy through 3D models and intuitive VR interactions.

The application is built using the XR Interaction Toolkit, which enables natural user interactions such as grabbing, rotating, and inspecting anatomical structures. Users can navigate through different modes and interact with objects using VR controllers.

The system architecture consists of several core modules.

2.1 Learning Mode

In this mode, users are guided through anatomical structures in a structured learning flow. The system presents anatomical parts along with visual explanations and contextual information. Additionally, checkpoint-based mini questions are included at certain stages to reinforce understanding. This combination ensures both informative and interactive learning.

2.2 Free Explore Mode

This mode allows users to freely interact with anatomical models without guidance. Users can grab and examine objects, and optionally display the names of the selected structures. Unlike the learning mode, detailed information panels are hidden to encourage self-exploration.

2.3 Quiz System

The quiz module is designed to evaluate the user's knowledge. It includes categorized questions, a countdown system before the quiz starts, and interactive answer selection within the VR environment.

2.4 AI Interaction System

The system includes an AI-assisted interaction module that enables users to communicate with an intelligent assistant. This module integrates speech-to-text (STT) and text-to-speech (TTS) functionalities, allowing users to ask questions verbally and receive spoken responses. The AI backend is supported by a Retrieval-Augmented Generation (RAG) approach, providing context-aware answers based on anatomical data.

2.5 Scene and Navigation System

The application includes a structured navigation system that manages transitions between different scenes and modes. A centralized state management approach is used to maintain user context, such as selected modules and interaction modes.

3. Implemented Features

This section describes the core features implemented in the VR Anatomy system, focusing on interaction design, visual feedback mechanisms, and system functionality.

3.1 Object Interaction System

The system allows users to interact with anatomical models using VR controllers through the XR Interaction Toolkit. Objects in the scene are configured as interactable components, enabling users to grab, move, and inspect them in a natural way.

When a user selects an object, it can be freely rotated and examined from different angles, providing a more immersive understanding of the anatomical structure. This interaction model enhances spatial awareness and supports experiential learning.

3.2 Visual Filtering System (Context-Based Visibility)

A context-based visual filtering system is implemented to improve clarity during the Free Explore mode. When a specific anatomical unit is selected, only the models related to that unit remain fully visible with their original appearance.

All unrelated anatomical structures are visually de-emphasized by reducing their opacity, creating a semi-transparent effect. This allows users to focus on the relevant components without completely removing the surrounding context.

Unlike traditional highlight systems, no visual enhancement is applied when an object is selected or grabbed. Instead, the system emphasizes contextual relevance by distinguishing related and unrelated structures through transparency.

This approach maintains the visual integrity of anatomical models while providing a clear and intuitive focus mechanism for the user.

3.3 Free Explore Mode Implementation

The Free Explore mode allows users to interact with anatomical models without guided instructions, providing a self-directed learning experience.

Users can freely grab, rotate, and inspect anatomical structures in a virtual environment. To keep the interface minimal and avoid visual clutter, object names are not displayed by default. Instead, an optional UI panel is provided.

Users can toggle the visibility of the object name panel using a button on the VR controller. When activated, the system displays the name of the currently held object. This ensures that users can access information when needed, without interrupting the immersive experience.

This design balances usability and immersion by allowing on-demand information access rather than constant visual overlays.

3.4 Quiz System Implementation

The Quiz System is designed as an adaptive assessment mechanism to evaluate the user's knowledge and guide their learning process.

Users access the quiz from the main menu and select a specific anatomical category (e.g., Circulatory System or Musculoskeletal System). Before the quiz begins, an informational screen is displayed, explaining the rules such as time limits, question types, and the countdown process.

Each quiz session includes:

- Multiple-choice questions
- Matching-type questions
- A time constraint (e.g., 30 seconds per question)
- A countdown sequence before the quiz starts

Questions are internally categorized into different difficulty levels (easy, medium, hard). The system dynamically adjusts the difficulty based on the user's performance:

- If the user answers correctly, subsequent questions become more difficult
- If the user struggles, the system provides easier questions

This adaptive mechanism allows the system to evaluate the user more accurately according to their level.

At the end of the quiz, the system analyzes the user's performance and provides feedback based on topic-wise accuracy. Based on this analysis, users are redirected to the relevant learning modules where they have weaknesses.

This creates a feedback loop between assessment and learning, making the system both evaluative and instructive.

3.5 AI-Based Interactive Learning System

An AI-based interactive learning system is integrated into the application to enhance user engagement and provide personalized educational support.

The system utilizes a Retrieval-Augmented Generation (RAG) architecture combined with the Gemini API to generate accurate and context-aware responses. Anatomical knowledge is stored in a structured dataset, and relevant information is retrieved and provided to the AI model to ensure reliable answers.

Users can interact with the AI system through both text and voice-based interfaces:

- Text-based interaction: Users can type questions and receive generated responses
- Voice-based interaction: Speech-to-Text (STT) is used to convert user speech into text, and Text-to-Speech (TTS) is used to deliver responses audibly

This allows for a more natural and immersive communication experience within the VR environment.

The AI system is designed to answer questions related to anatomical structures, explain concepts, and support the learning process dynamically. By combining RAG with multimodal interaction (text + voice), the system provides both accessibility and flexibility for different user preferences.

4. Implementation Details

This section presents the development process of the VR Anatomy project, including versioning, iteration workflow, repository reference, and system outputs.

4.1 Versioning

The project was developed using an internal versioning approach to track the progression of system features. Instead of formal public releases, versioning was used conceptually to represent major development milestones.

The versioning structure reflects the internal development stages of the system:

- **v0.1** – Initial VR setup and basic interaction system
- **v0.2** – Early Learning Mode structure
- **v0.3** – Free Explore mode and visual filtering system
- **v0.4** – Quiz system with adaptive difficulty
- **v0.5** – AI integration (RAG + Gemini API + STT/TTS)
- **v1.0** – Final integrated system (in progress)

4.2 Iteration Log

The development process followed an iterative and non-linear approach. Different components of the system were developed in parallel and continuously refined.

Instead of a strictly sequential workflow, the team worked on multiple modules simultaneously, including interaction systems, AI integration, and user interface design.

The iterations can be conceptually summarized as follows:

- **Iteration 1:** Initial VR setup and core interaction system
- **Iteration 2:** Learning Mode development
- **Iteration 3:** Free Explore mode and context-based visual filtering
- **Iteration 4:** Quiz system with adaptive difficulty mechanism
- **Iteration 5:** AI-based system integration (RAG + Gemini API + voice interaction)
- **Iteration 6:** UI improvements, optimization, and ongoing refinements

This flexible development approach allowed continuous testing, feedback integration, and progressive improvement of the system.

4.3 Repository Reference

The project source code is maintained in a GitHub repository:

- https://github.com/CankayaUniversity/ceng-407-408-2025-2026-VR-Anatomy-VR-Based-Educational-Interactive-Human-Anatomy-Training-Platform

The repository includes:

- Unity project files
- AI backend components
- Version history and commit records
- Supporting resources and documentation

4.4 Screenshots

Screenshots are provided to demonstrate the implemented features and user interface of the VR Anatomy system. The images are organized according to the user flow within the application.

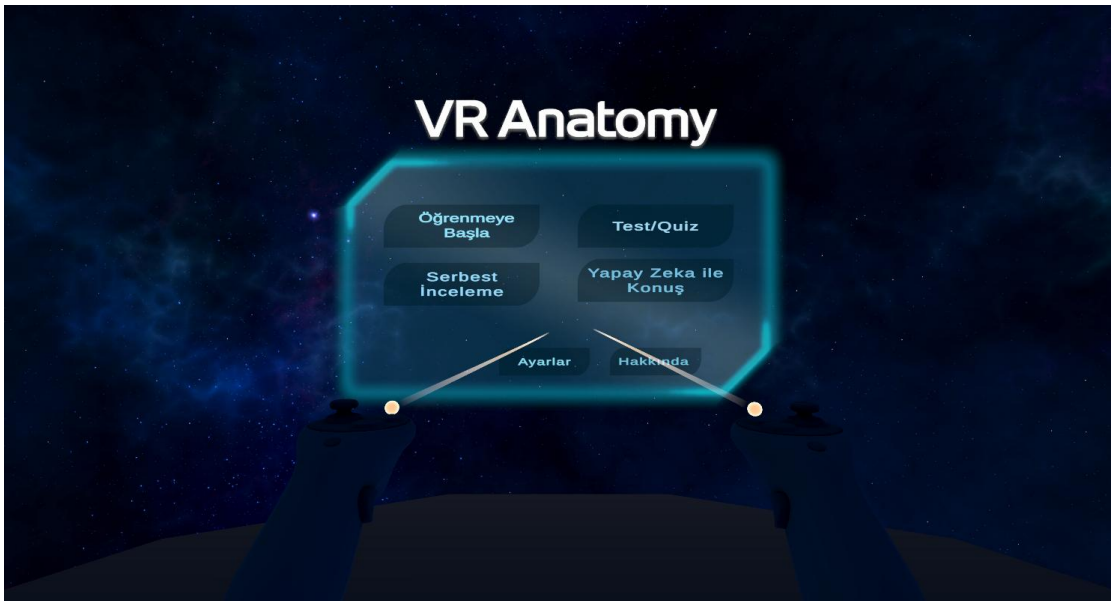

Figure 1: Main Menu Interface

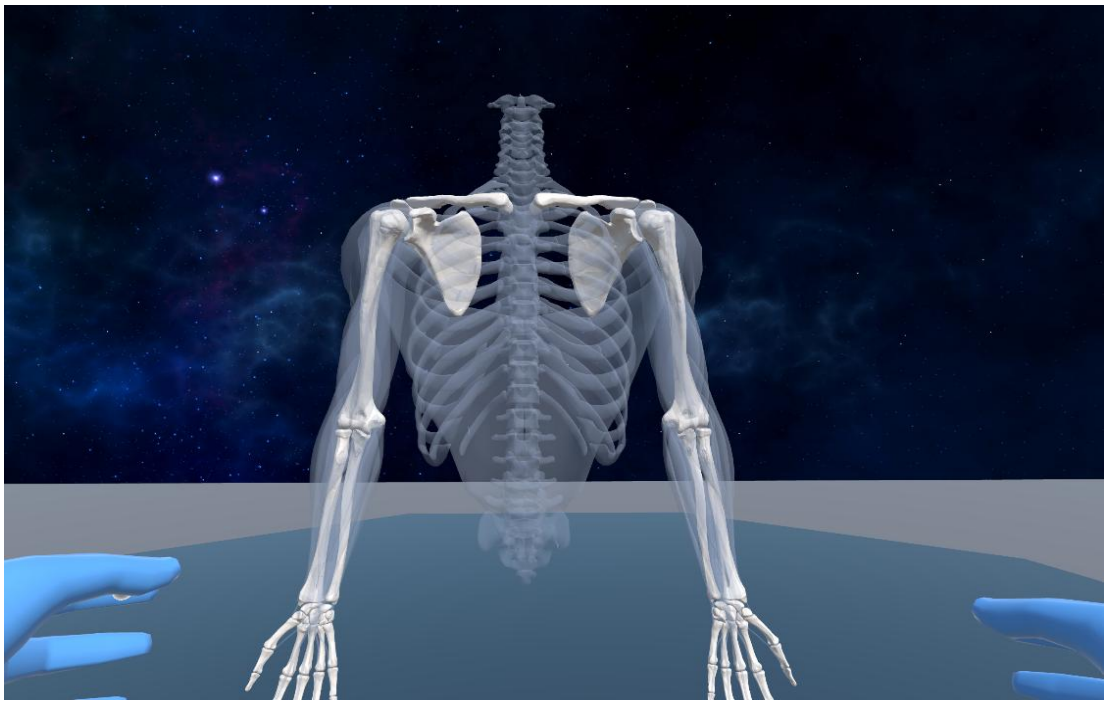

Figure 2: Free Explore Mode with Visual Filtering

Figure 3: Free Explore Mode with Object Interaction and UI Panel

Figure 4: Quiz Information Screen

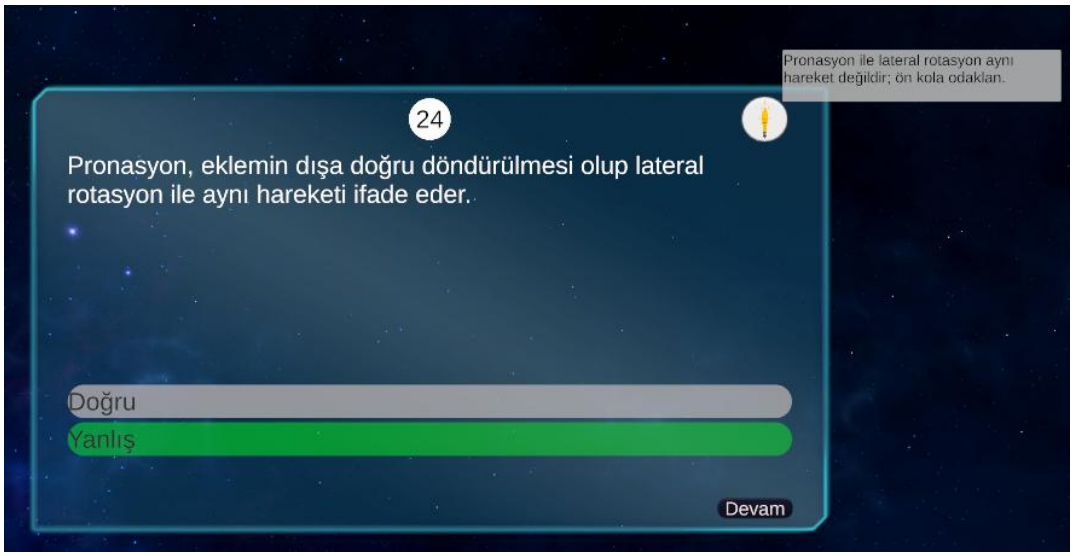

Figure 5: Quiz Mode Interface (True/False Question)

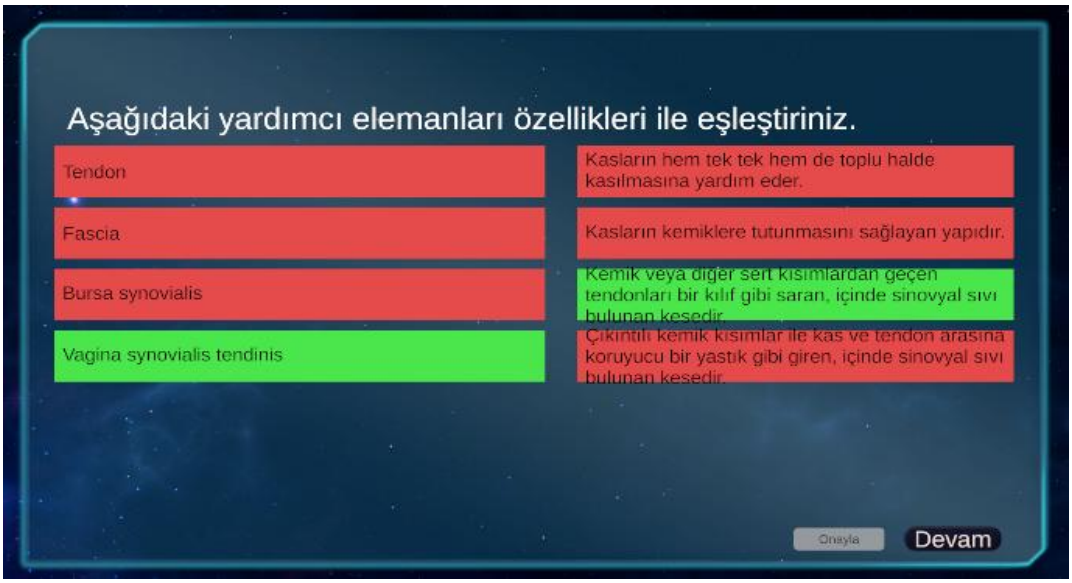

Figure 6: Quiz Mode Interface (Matching Question)

Figure 7: Instructional UI Panel (User Guidance System)

Figure 8: Learning Mode Interface

Figure 9: AI-Based Interaction Interface